

Monetary–Fiscal Coordination as a State-Contingent Dynamic Game: Steady States, Shocks, Commitment, and an Australian Calibration

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Research draft — September 26, 2026

Abstract

This paper develops a monetary–fiscal policy game calibrated to Australian data. It keeps a single social-welfare criterion fixed while allowing the central bank and fiscal authority to place different relative weights on inflation and the output gap. We first solve cooperative, simultaneous Nash, and Stackelberg versions of a finite-horizon open-loop game. With no shock and zero gap targets, every solution has the same zero steady state; demand and cost-push shocks activate large policy-mix externalities. We then solve a recursive linear–quadratic game in which output, inflation, debt, and lagged instruments are states, and policies are stationary Markov feedback rules. Adjustment costs make current instruments investments in future strategic positions. In a first institutional stage, each authority delegates an adjustment coefficient to its future policymaker; in the second stage, the policymakers play the feedback Nash game after shocks. The Australian feedback equilibrium is stable and strongly history dependent. The institutional Nash choice, $(\lambda_M, \lambda_F) = (0.05, 0.04)$, equals the primitive adjustment weights, while the social design is $(0.05, 0.115)$. Thus the baseline does not manufacture a commitment arms race: it shows under-provision of fiscal inertia. Recursive Nash raises social loss by only 0.49 per cent after a demand shock and 0.06 per cent after a cost-push shock, much less than the open-loop benchmark. The difference is economically informative but not a clean horse race because the recursive implementation uses an estimated semi-structural timing closure. Open-loop commitment, Markov discretion, and institutional design answer different questions and should be reported separately.

Keywords: monetary–fiscal coordination; dynamic games;

Markov-perfect equilibrium; strategic complementarity; fiscal dominance; Australia.

JEL codes: E52, E58, E61, E62, C73.

1 Introduction

Monetary and fiscal authorities act on the same economy but need not evaluate the resulting inflation, activity, debt, or instrument movements in the same way. This observation is enough to create a policy game even when both institutions are benevolent and the model contains no political distortion. It does not, however, imply that every shock produces the same kind of game. Whether the instruments are strategic complements or substitutes, whether an authority free rides, and whether leadership improves outcomes depend on the shock, nominal rigidity, the state variables, and the timing of commitment.

The motivating draft distinguishes two mechanisms. Following a demand shock, both authorities want to stabilise activity but may try to shift the adjustment cost to the other authority—a “GFC effect.” Following a supply disturbance, the central bank’s concern for inflation can produce tightening while the fiscal authority’s concern for activity produces expansion. The policies then offset one another at the macroeconomic level while becoming mutually reinforcing in their numerical instrument settings—a “post-COVID effect.” The draft also makes prior policy a state through adjustment costs and asks whether institutions deliberately acquire commitment technology. These are recursive, state-contingent questions.

The first computational block answers a narrower but useful question. At the beginning of a finite horizon, after a shock path is known, the authorities choose complete instrument paths. This open-loop game isolates the policy-mix externality and provides a benchmark. The new recursive block instead lets authorities reoptimise at every observed state and makes today’s instruments tomorrow’s lagged-policy states. A first-stage institutional game then lets each authority choose the adjustment coefficient delegated to its future self.

This revision makes that boundary explicit. It contributes five results. First, it embeds fiscal policy and debt in a transparent extension of the three-equation New Keynesian model. Second, it separates agency mandates from true social welfare. Third, it establishes experimentally that the calibrated no-shock steady state contains no realised conflict, while shocks generate large policy-game losses in the open-loop benchmark. Fourth, it solves the recursive Markov-perfect game requested in the original draft. Fifth, it solves a two-stage institutional game in delegated adjustment costs and shows that, at the Australian baseline, decentralised authorities retain their primitive inertia while a social designer chooses substantially more fiscal inertia. Open-loop Nash and leadership remain benchmarks rather than final equilibrium concepts.

2 From three equations to a monetary–fiscal game

2.1 The private-economy block

Let x_t denote the output gap, π_t inflation relative to target, i_t the monetary-policy instrument relative to neutral, and g_t a fiscal-demand instrument, where $g_t > 0$ is expansionary. A compact forward-looking model is

$$x_t = \mathbb{E}_t x_{t+1} - \sigma(i_t - \mathbb{E}_t \pi_{t+1}) + \psi g_t + d_t, \quad (1)$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t, \quad (2)$$

$$b_{t+1} = \rho_b b_t + \chi_g g_t + \chi_i i_t - \chi_\pi \pi_t. \quad (3)$$

Here d_t is a demand disturbance, u_t is a cost-push disturbance, and b_t is the public-debt gap. The signs match the code: a larger i_t is tighter and a larger g_t is more expansionary. Equation (1) is the IS curve, (2) is the New Keynesian Phillips curve, and (3) is a deliberately parsimonious debt law.

It is potentially misleading to call this merely a “four-equation model.” The conventional three-equation model consists of IS, Phillips, and an exogenous monetary rule. Once both policies are optimised, the Taylor rule is replaced by the central bank’s first-order condition and a fiscal first-order condition is added. With debt retained, the equilibrium has five blocks: IS, Phillips, debt accumulation, monetary optimality, and fiscal optimality. For active–passive

regime comparisons we also use the explicit rule closure

$$i_t = \rho_i i_{t-1} + (1 - \rho_i)(\phi_\pi \pi_t + \phi_x x_t) + \varepsilon_t^i, \quad (4)$$

$$g_t = \rho_g g_{t-1} + (1 - \rho_g)(\gamma_b b_t + \gamma_x x_t) + \varepsilon_t^g. \quad (5)$$

Equations (4)–(5) are an alternative closure, not additional behavioural assumptions inside the optimising game.

2.2 One social objective, two agency mandates

The primitive social loss is fixed across every game:

$$\begin{aligned} \mathcal{L}^S = \frac{1}{2} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t & [\omega_\pi^S \pi_t^2 + \omega_x^S x_t^2 + \omega_b^S b_t^2 + \omega_i^S i_t^2 + \omega_g^S g_t^2 \\ & + \omega_{\Delta i}^S (i_t - i_{t-1})^2 + \omega_{\Delta g}^S (g_t - g_{t-1})^2]. \end{aligned} \quad (6)$$

It is the welfare yardstick and therefore does not change when the strategic regime changes. Cooperation means that a single planner minimises (6); it does not mean adding together two differently weighted agency losses.

The agency losses have the same form but different inflation and output-gap weights:

$$\mathcal{L}^j = \frac{1}{2} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\omega_\pi^j \pi_t^2 + \omega_x^j x_t^2 + \omega_b^j b_t^2 + \omega_i^j i_t^2 + \omega_g^j g_t^2 + \omega_{\Delta i}^j (\Delta i_t)^2 + \omega_{\Delta g}^j (\Delta g_t)^2], \quad (7)$$

where $j \in \{M, F\}$. In the baseline calibration the monetary authority is more inflation averse, $(\omega_\pi^M, \omega_x^M) = (1.5, 0.25)$, while the fiscal authority is more output oriented, $(\omega_\pi^F, \omega_x^F) = (0.5, 0.75)$. All other weights are held fixed. The social weights are $(\omega_\pi^S, \omega_x^S) = (1, 0.5)$. Thus the experiment changes mandates, not technology or the definition of welfare.

This distinction also explains why cooperation can involve a larger response by one instrument than a decentralised equilibrium. Cooperation internalises both instruments' effects and assigns the policy mix according to (6). Nash policymakers instead minimise different losses and may partially offset or free ride on one another. A larger cooperative movement is not generated by changing the planner's weights across cases; it is an equilibrium allocation of a fixed social objective.

3 Timing, states, and equilibrium concepts

3.1 Shocks and the strategic state

For the intended stochastic model, take

$$d_{t+1} = \rho_d d_t + \varepsilon_{t+1}^d, \quad u_{t+1} = \rho_u u_t + \varepsilon_{t+1}^u, \quad (8)$$

with innovations not known at t . The implemented recursive state is

$$z_t = (x_{t-1}, \pi_{t-1}, b_t, i_{t-1}, g_{t-1}, d_t, u_t)'. \quad (9)$$

Lagged instruments matter because adjustment costs make past policy affect the current marginal cost of action. Debt transmits fiscal and monetary choices across time. Persistent shocks change the distribution of future states. Lagged output and inflation implement

the Australian semi-structural timing closure described below. A fully forward-looking New Keynesian implementation would instead augment the state with equilibrium expectational variables and, under Ramsey commitment, promised inflation or planner multipliers.

The timeline is: the state z_t is observed; private agents form rational expectations using anticipated policy functions; the authorities choose their instruments according to the equilibrium timing; outcomes occur; and new innovations produce z_{t+1} . This timing makes the policies contingent on the realised state, not just on calendar time.

3.2 Feedback Nash and Markov perfection

Let the strategy functions be

$$i_t = F_M(z_t), \quad g_t = F_F(z_t). \quad (10)$$

A stationary Markov-perfect Nash equilibrium is a pair (F_M, F_F) such that, given the other authority's policy function and rational private expectations, each authority solves its Bellman problem at every state. Schematically,

$$V_M(z) = \min_i \{ \ell_M(z, i, F_F(z)) + \beta \mathbb{E}[V_M(z') \mid z, i, F_F(z)] \}, \quad (11)$$

$$V_F(z) = \min_g \{ \ell_F(z, F_M(z), g) + \beta \mathbb{E}[V_F(z') \mid z, F_M(z), g] \}. \quad (12)$$

In the linear-quadratic approximation, value functions are quadratic and the policy functions are linear, $i_t = -K_M z_t$ and $g_t = -K_F z_t$. The equilibrium is obtained from coupled Riccati equations, with stabilising and determinacy conditions imposed on the joint private-policy system.

To see the investment channel, write the recursive transition as $z' = Az + B_M i + B_F g$. Given quadratic continuation matrices P_M and P_F , the simultaneous policy first-order conditions have the form

$$\begin{bmatrix} Q_{ii}^M + \beta B_M' P_M B_M & Q_{ig}^M + \beta B_M' P_M B_F \\ Q_{gi}^F + \beta B_F' P_F B_M & Q_{gg}^F + \beta B_F' P_F B_F \end{bmatrix} \begin{bmatrix} i_t \\ g_t \end{bmatrix} = - \begin{bmatrix} Q_{iz}^M + \beta B_M' P_M A \\ Q_{gz}^F + \beta B_F' P_F A \end{bmatrix} z_t. \quad (13)$$

The vectors B_M and B_F contain unit loadings that make i_t and g_t next period's lagged instruments. Hence the continuation terms in (13) are the formal strategic-investment motive: a current instrument changes future macro states, future adjustment costs, and the other authority's future feedback action.

This is the natural main solution concept for the original draft. The authority that deviates today understands how the deviation changes debt and the next period's adjustment-cost state, but it cannot bind its future self to ignore information that will later be available. Consequently the solution speaks directly to credibility, reoptimisation, and policy used as a strategic state.

3.3 What open-loop equilibrium does and does not mean

The open-loop numerical block instead selects complete paths $I = (i_0, \dots, i_{T-1})'$ and $G = (g_0, \dots, g_{T-1})'$ at date zero. Conditional on a known shock path, the linear economy can be written

$$q = q^0 + A_I I + A_G G, \quad (14)$$

where q stacks macroeconomic outcomes. Quadratic agency losses imply affine best responses

$$I = a_M + B_M G, \quad G = a_F + B_F I. \quad (15)$$

The open-loop Nash path solves the two equations simultaneously. The round-trip matrix $B_M B_F$ measures the strength of iterative reaction, and $\rho(B_M B_F) < 1$ is a useful local convergence diagnostic.

Open-loop equilibrium is appropriate if the economic experiment is explicitly one of full path commitment after the shock is observed and no unanticipated innovation subsequently arrives. It is also valuable as a transparent benchmark. It is not sufficient when the object of interest is policy under recurring shocks, discretion, or strategic investment in adjustment costs. The path is time-indexed rather than state-contingent, and an authority does not reoptimise after histories that were not anticipated at date zero.

Nor does the spectral radius alone identify strategic complementarity. It is unsigned and summarises a high-dimensional round trip. Classification requires the signed cross-derivatives $\partial i_t / \partial g_s$ and $\partial g_t / \partial i_s$, evaluated at an economically specified state or shock. With the present sign convention, positive contemporaneous slopes are complementarity in instrument levels: fiscal expansion elicits monetary tightening, and tighter money elicits more fiscal expansion. The two policies then offset one another in aggregate demand.

3.4 Leadership and the commitment meta-game

The existing Stackelberg calculations are also open loop. A leader chooses a complete path while internalising the follower's complete-path best response. This can make monetary and fiscal leadership look similar because both leaders have nearly the same ability to exploit the same linear reaction mapping and because the calibration is close to a common welfare problem once the follower is internalised.

The recursive counterpart is feedback Stackelberg equilibrium. There the leader anticipates the follower's state-contingent response at every future state. The original draft additionally suggests a prior institutional stage: before shocks are realised, authorities choose commitment or adjustment-cost parameters, say λ_M and λ_F ; after shocks, they play the feedback policy game. Formally, the ex ante problem is

$$\lambda_j^* \in \arg \min_{\lambda_j} \mathbb{E}[V_j(z_0; \lambda_j, \lambda_{-j})], \quad (16)$$

where V_j is generated by the continuation MPE. In the implementation, the delegated coefficient replaces the own-instrument adjustment coefficient in the future policymaker's operational problem, but the stage-one authority evaluates the outcome using its original mandate. Its institutional objective is

$$\mathcal{J}_j(\lambda_j, \lambda_{-j}) = \mathbb{E} \mathcal{L}_{\text{primitive}}^j(F_M(\lambda), F_F(\lambda)) + c_\lambda(\lambda_j - \bar{\lambda}_j)^2, \quad (17)$$

where $\bar{\lambda}_j$ is the primitive adjustment weight. Centering the implementation cost at the primitive separates strategic delegation from an arbitrary mechanical preference for zero inertia. A constitutional planner chooses both coefficients using (6) and the sum of implementation costs. This separates leadership within stabilisation from the acquisition of commitment technology.

4 The steady state and the shock-contingent type of game

4.1 No-shock steady state

Set $d = u = 0$ and use zero gap targets. Equations (1)–(3) and either policy closure admit

$$x^* = \pi^* = b^* = i^* = g^* = 0. \quad (18)$$

Every term in each quadratic loss is then zero. Because policy costs are strictly convex, no authority can improve on this allocation by moving its instrument. The numerical zero-shock experiment verifies that cooperation, Nash, monetary leadership, and fiscal leadership all return exactly this steady state.

This finding should be stated carefully. It does not say that institutions or coordination are irrelevant in normal times. Reaction slopes and the welfare consequences of future states remain latent, and an ex ante commitment choice can still matter because shocks occur with positive probability. It says that the present aligned-target calibration does not generate a realised steady-state policy conflict. A non-zero steady-state game would require an additional wedge such as different inflation targets, a fiscal spending bias, distortionary tax objectives, a debt target conflict, or seigniorage pressure.

Table 1: Numerical no-shock steady state

Strategy	social loss	$\max x $	$\max \pi $	$\max i $	$\max g $
Cooperation	0	0	0	0	0
Simultaneous Nash	0	0	0	0	0
Monetary leader	0	0	0	0	0
Fiscal leader	0	0	0	0	0

4.2 A taxonomy by shock

The shock determines which disagreement is economically active. Table 2 gives the disciplined interpretation intended for the full feedback model.

Table 2: Shock-contingent strategic interaction

State	Main policy trade-off	Likely strategic mechanism	Empirical diagnostic
No shock, aligned targets	None realised at the zero-gap steady state	Dormant reaction structure; coordination matters ex ante	Identical zero policies and losses
Demand contraction	Both agencies favour expansion, but dislike changing their own instruments	Burden sharing and free riding; stabilisation efforts can be substitutes	Signed cross-responses and each authority's contribution to closing the output gap
Cost-push inflation	Inflation stabilisation conflicts with output stabilisation	Monetary tightening and fiscal expansion can be complements in instrument levels but offsets in demand	Signed policy slopes, policy-mix distance, inflation/output outcomes
Debt or solvency shock	Price stability, debt stabilisation, and valuation effects interact	Active-passive regimes and possible fiscal dominance	Determinacy, debt backing, and nominal-liability valuation

Nominal rigidity is pivotal after a cost-push shock. When κ is finite, lower inflation requires an output sacrifice. Because the monetary authority places greater weight on inflation and the fiscal authority places greater weight on output, their preferred points on that trade-off differ. The fiscal response then raises the central bank's desired rate, while the monetary response raises the fiscal authority's desired support. This is precisely the case in which feedback, expectations, and commitment may alter the result most.

5 Australian calibration and computation

The code is written entirely in Julia. The descriptive IS and Phillips coefficients are estimated using quarterly Australian data over 1993Q1–2019Q4, with public series cached alongside checksums for reproducibility. The sample ends before the pandemic so that the cost-push experiment is not mechanically fit to the episode it is intended to illuminate. The debt block and welfare weights are transparent calibrations rather than claimed structural estimates.

The open-loop benchmark retains equations (1)–(3). The recursive experiment uses the directly estimable semi-structural timing

$$x_t = \rho_x x_{t-1} - \sigma(i_t - \pi_{t-1}) + \psi g_t + d_t, \quad (19)$$

$$\pi_t = \rho_\pi \pi_{t-1} + \kappa x_t + u_t, \quad (20)$$

$$b_{t+1} = \rho_b b_t + \chi_g g_t + \chi_i i_t - \chi_\pi \pi_t. \quad (21)$$

Current shocks are observed before policy. The controls affect current output, output affects current inflation, and current inflation affects end-of-period debt. Equations (19)–(21), lagged instruments, and shock laws form a linear transition $z_{t+1} = Az_t + B_M i_t + B_F g_t + \varepsilon_{t+1}$. This closure is recursive and empirically transparent, but it is not identical to the forward-looking private-sector block; the two sets of welfare numbers must therefore not be interpreted as a controlled comparison of solution concepts alone.

Table 3: Selected baseline parameters

Parameter	Value	Interpretation
β	0.9900	quarterly discount factor
σ	0.03695	semi-elasticity to the real-rate gap
ψ	0.05142	public-demand multiplier in the IS equation
κ	0.15511	Phillips-curve slope
ρ_x	0.62875	output-gap persistence in recursive model
ρ_π	0.56255	inflation persistence in recursive model
ρ_b	0.995	debt persistence
$(\omega_\pi^S, \omega_x^S, \omega_b^S)$	(1, 0.5, 0.15)	social weights
$(\omega_\pi^M, \omega_x^M)$	(1.5, 0.25)	monetary mandate
$(\omega_\pi^F, \omega_x^F)$	(0.5, 0.75)	fiscal mandate

For each deterministic shock path, the program constructs the linear path map (14), derives exact quadratic best responses, and solves four regimes: the social planner, simultaneous Nash, monetary leadership, and fiscal leadership. It also varies mandate divergence and policy adjustment costs, and solves the explicit rule system in monetary- and fiscal-dominance analogues. For the recursive experiment, coupled algebraic Riccati iterations solve the feedback Nash value functions. A grid over delegated adjustment coefficients $[0, 0.15]^2$ with spacing 0.005 solves the first-stage game; the implementation cost is $c_\lambda = 0.005$.

6 Results

6.1 Shocks activate the coordination problem

Figure 2 scales the demand and cost-push innovations from zero to 1.5 sample standard deviations. At zero, the Nash welfare gap and the distance between cooperative and Nash

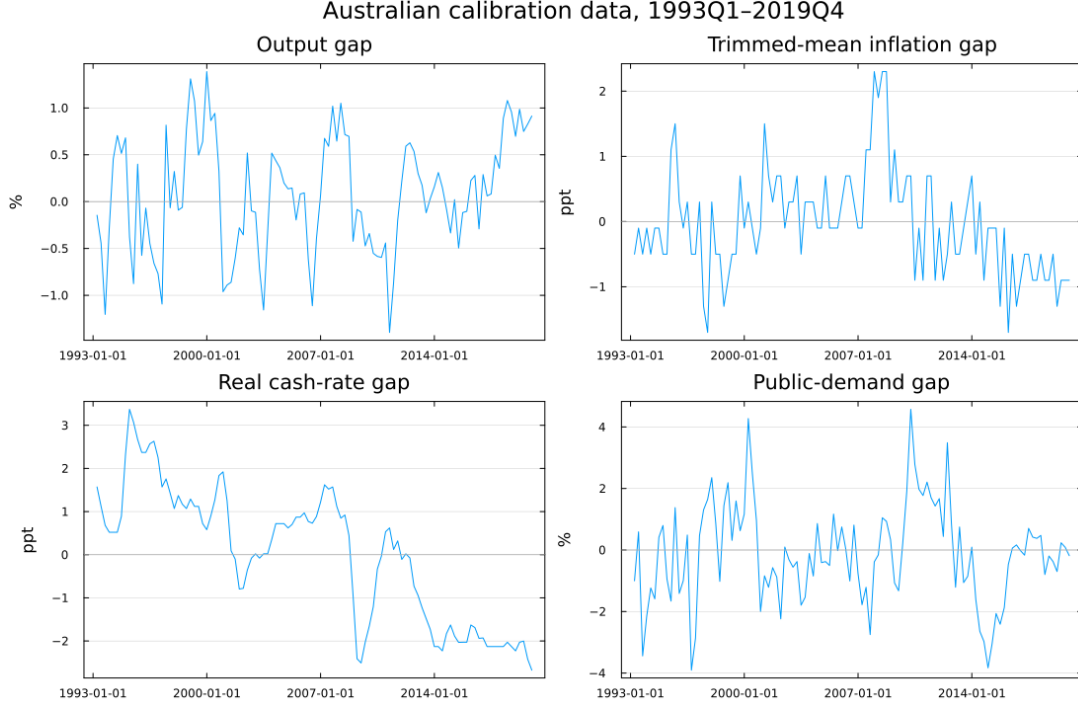


Figure 1: Australian data used for the descriptive calibration.

policy paths are both zero. As shocks grow, the path distance rises linearly and excess Nash loss rises quadratically, as expected in a linear-quadratic model. The supply-shock curves are steeper: the shock activates the disagreement over the inflation-output trade-off.

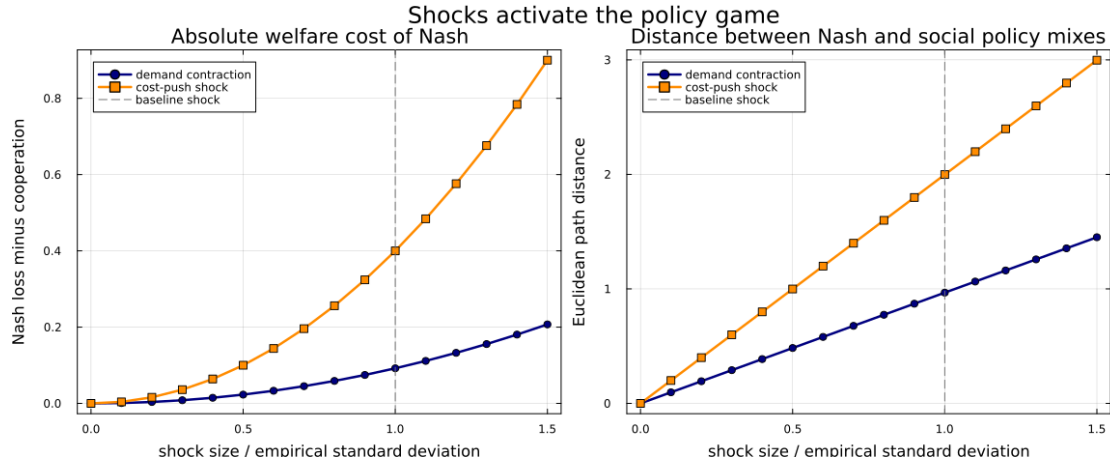


Figure 2: Shock activation of the coordination problem. The left panel is $\mathcal{L}_N^S - \mathcal{L}_C^S$; the right panel is the Euclidean distance between the Nash and cooperative instrument paths.

6.2 Demand contraction: burden sharing and free riding

After a one-standard-deviation contractionary demand shock, cooperative social loss is 0.0972 and simultaneous Nash loss is 0.1892, a 94.8 per cent increase. The peak monetary response is 0.067 under cooperation but 0.306 under Nash; the peak fiscal response is 0.075 under cooperation and 0.257 under Nash. The large gross movements do not indicate stronger stabilisation. They partly reflect the authorities reacting to and offsetting the policy mix

generated by the other authority. The output peaks remain similar, while debt accumulation is much larger under Nash.

The open-loop leadership outcomes are close to the planner: social loss is 0.0991 with monetary leadership and 0.0986 with fiscal leadership. This result shows that internalising the follower's reaction removes most of the static path externality. It should not yet be read as evidence that the institutional identity of the leader is unimportant. Feedback leadership may differ because debt and lagged instruments change subsequent bargaining positions, while an ex ante commitment choice changes continuation values before the shock occurs.

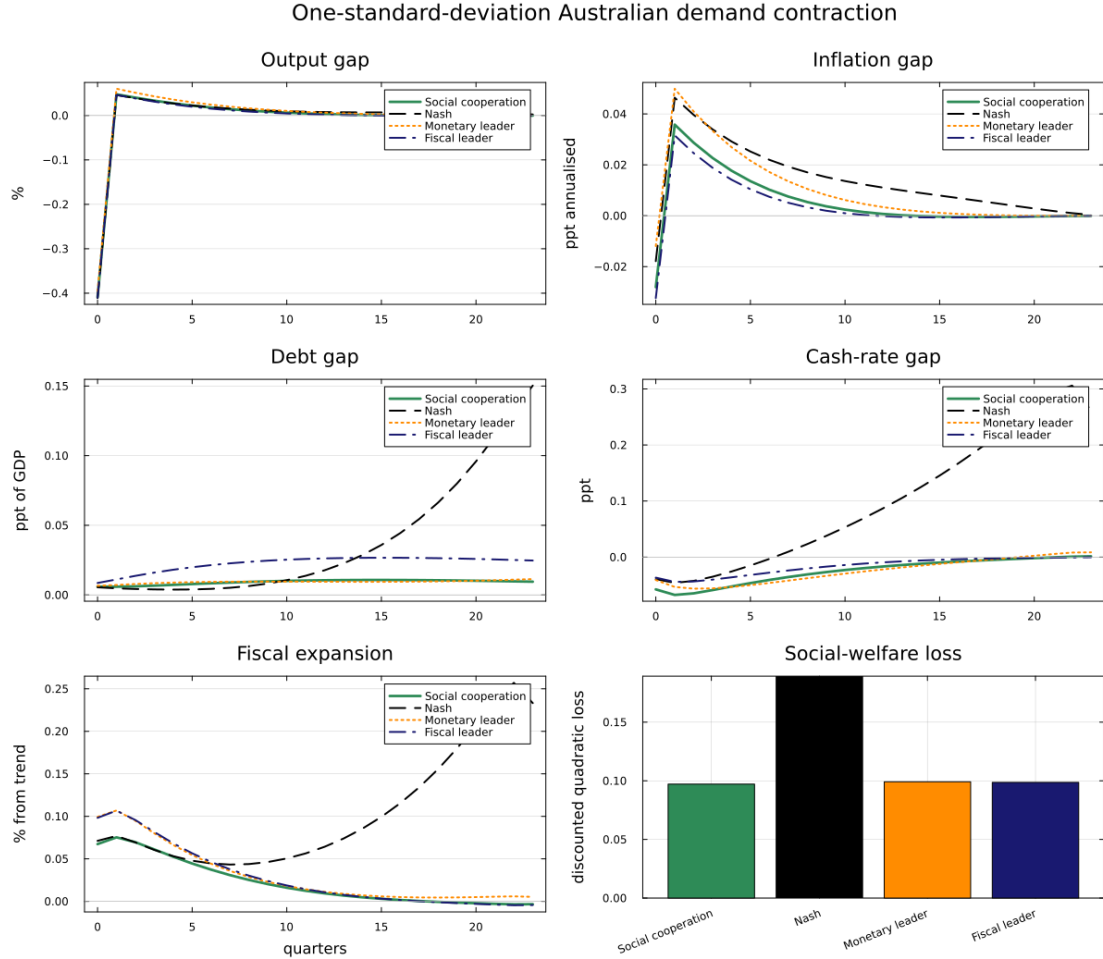


Figure 3: Responses to a contractionary demand shock under cooperation, simultaneous open-loop Nash, and open-loop leadership.

6.3 Cost-push shock: the offsetting-policy game

After a one-standard-deviation cost-push shock, cooperative social loss is 0.3762 and Nash loss is 0.7762, an increase of 106.3 per cent. Nash produces a peak monetary response of 0.607 and a peak fiscal response of 0.500, compared with 0.050 and 0.046 under cooperation. This is the clearest baseline example of strategic complementarity in instrument levels. The central bank tightens against inflation; the fiscal authority supports activity; each action raises the other's incentive to respond. Because i_t and g_t enter the IS curve with opposite signs, complementarity between the numerical instruments means macroeconomic offset, not common-direction stimulus.

Again, open-loop leadership nearly restores cooperative welfare: social loss is about 0.386 under either leader. The similarity follows from the common linear path technology and the leader's internalisation of the follower's full-path response. Whether it survives recursive reoptimisation is a substantive question for the feedback Stackelberg experiment.

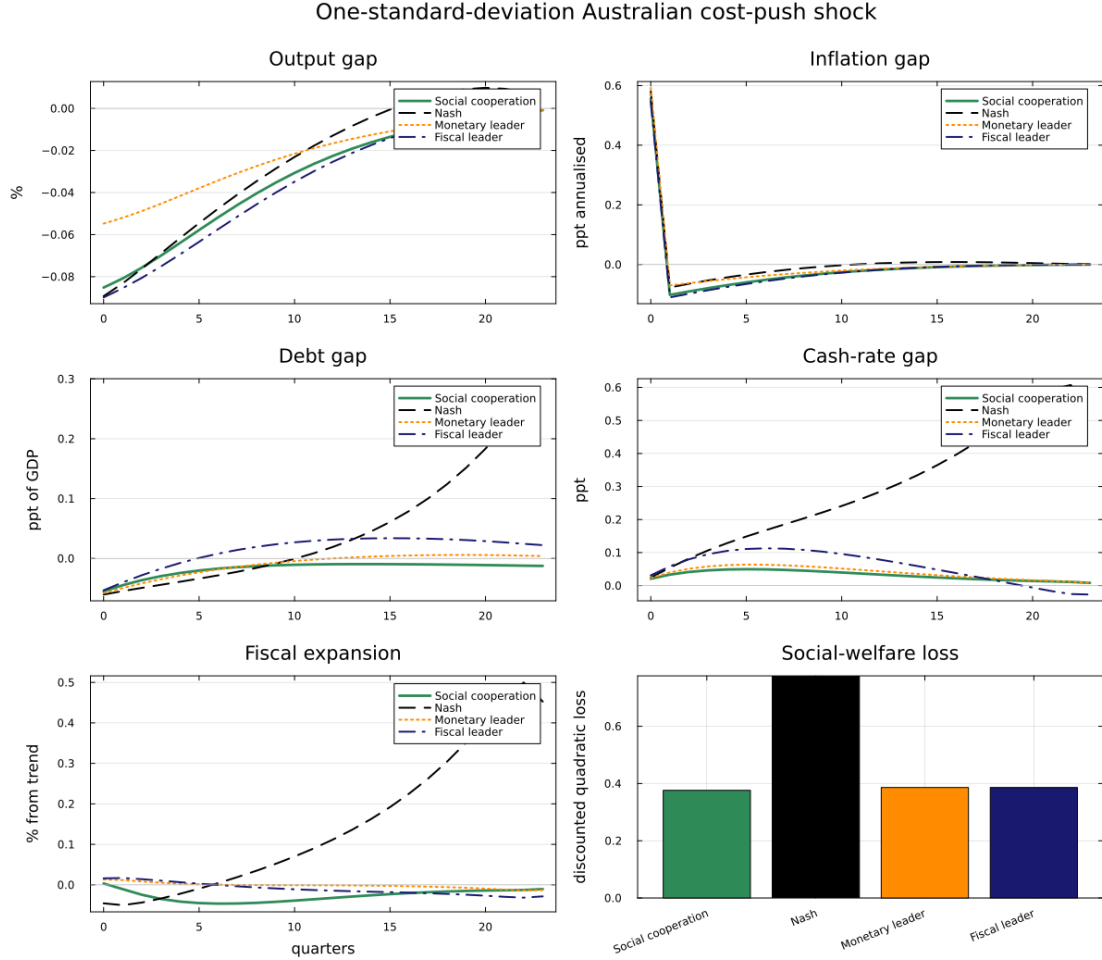


Figure 4: Responses to a cost-push inflation shock.

6.4 Mandate divergence and the coordination boundary

At the baseline mandate divergence, the monetary best-response norm is 1.313 and the fiscal norm is 0.742. The round-trip spectral radius is 0.950, close to the unit boundary. This explains why iterated reactions can amplify small differences and why decentralised policies are so much larger than cooperative policies. It does not establish that every element of both reaction matrices is positive, and it should not be used as a shock-invariant label for the game.

Figures 5 and 6 show that stronger mandate divergence and lower instrument costs move the open-loop system toward the coordination boundary. The economically important next step is to reproduce these maps using the eigenvalues of the closed-loop stochastic system and signed, state-contingent cross-responses.

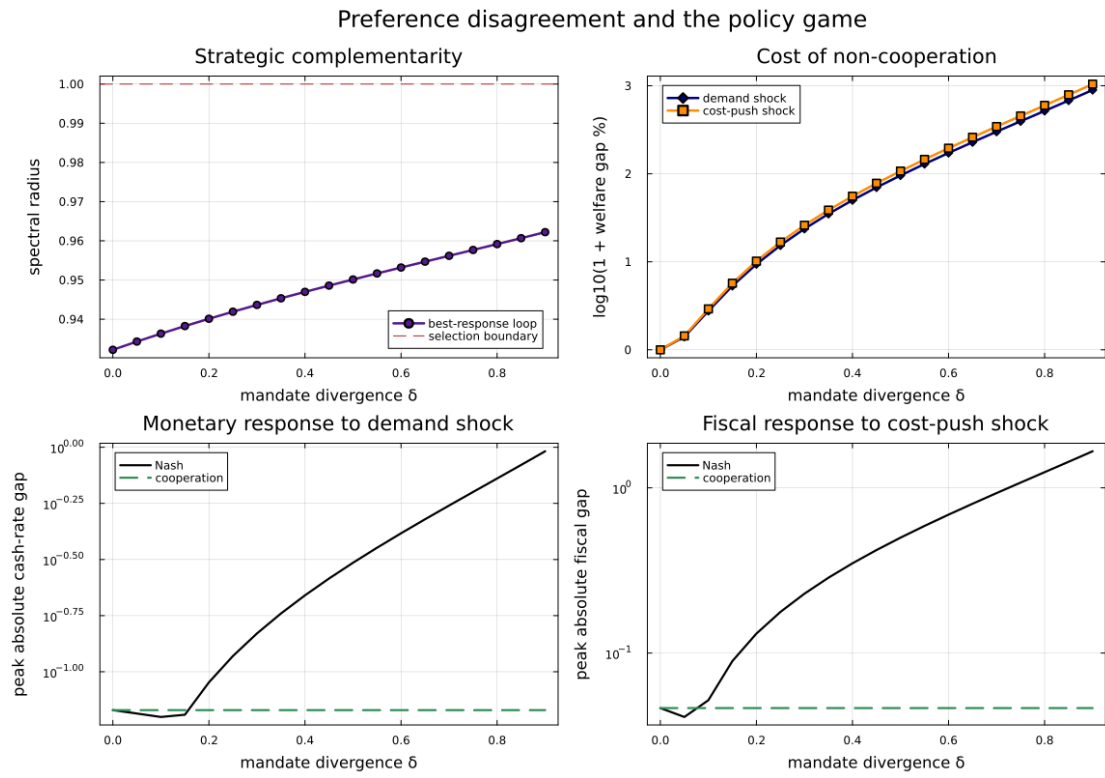


Figure 5: Open-loop reaction strength and welfare as agency mandates diverge.

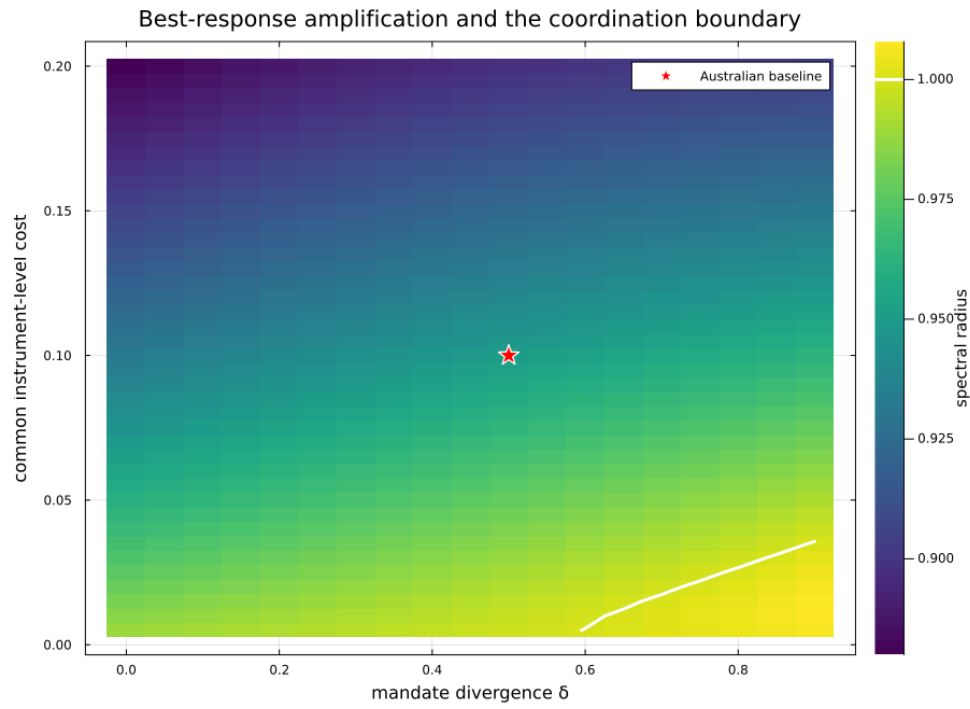


Figure 6: The open-loop coordination boundary across mandate divergence and policy-cost calibrations.

6.5 Does the model nest fiscal dominance?

The rule closure nests an active–passive regime comparison in the sense of [Leeper \(1991\)](#). In the monetary-dominance configuration, the interest-rate rule responds strongly to inflation and fiscal policy stabilises debt. In the fiscal-dominance analogue, monetary policy responds weakly and fiscal policy is less debt responsive. The latter produces peak inflation of 0.757 versus 0.404 and a peak policy rate of 0.244 versus 0.480 in the monetary-dominance case. Figure 7 displays the paths.

This is a nesting of the active–passive logic, not a complete fiscal theory of the price level. The parsimonious debt equation lets inflation reduce the real debt gap, which can make the calibrated quadratic social loss lower in the fiscal-dominance analogue. A full fiscal-dominance claim requires nominal government liabilities, a consolidated intertemporal budget constraint, explicit primary-surplus backing, and equilibrium price-level valuation. The current result should therefore be read as a regime diagnostic rather than a welfare endorsement.

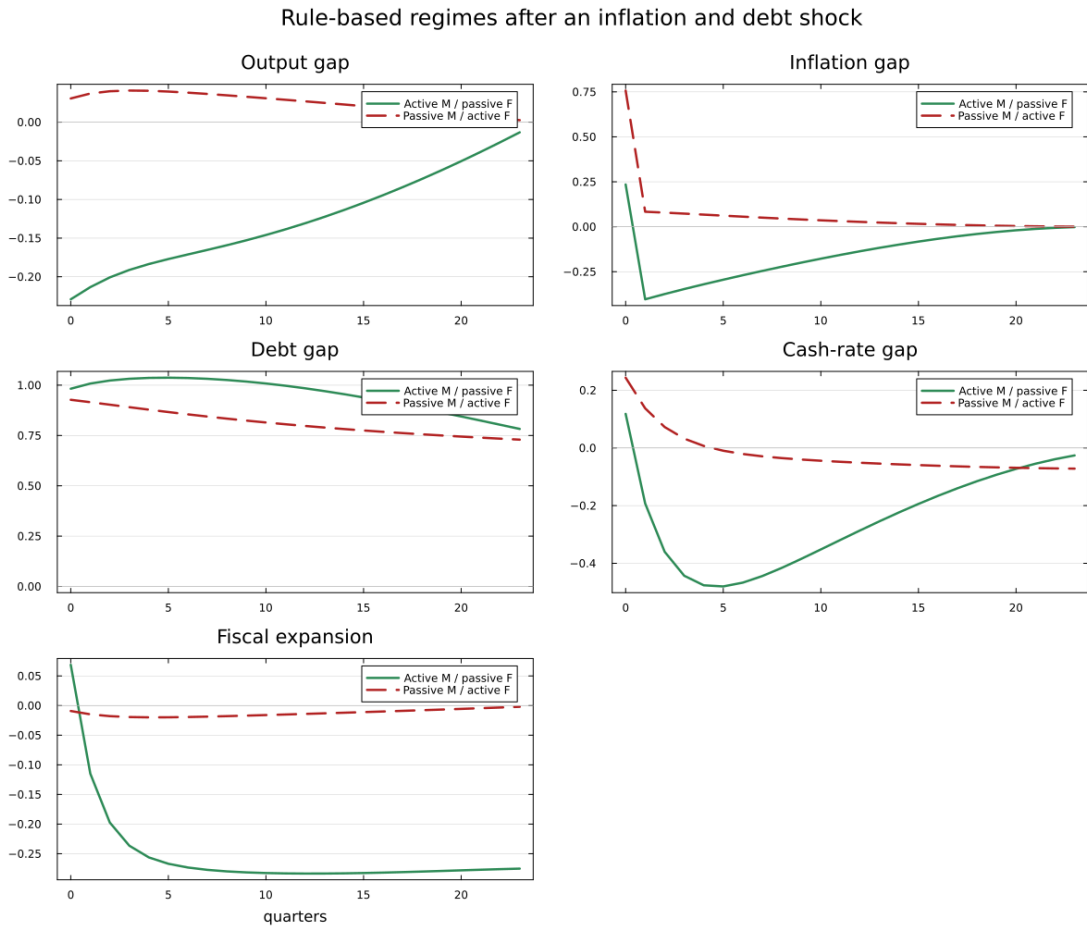


Figure 7: Active–passive rule-system comparison.

6.6 Recursive Markov-perfect equilibrium

The recursive feedback Nash calculation converges in 177 coupled Riccati iterations. Its closed-loop spectral radius is 0.9383, so the equilibrium is stable but persistent. The estimated rules provide direct evidence that adjustment costs make policy a strategic state. The monetary rule loads 0.265 on its own lagged instrument and the fiscal rule loads 0.217 on its own lagged

instrument. When both delegated adjustment coefficients are set to zero, these two history coefficients are exactly zero. Current policy therefore changes not only current demand and inflation, but the marginal cost and strategic position inherited by future policymakers.

Table 4 reports the main recursive results. The Nash welfare gap is 0.49 per cent after the demand contraction and 0.06 per cent after the cost-push shock. Removing policy inertia increases gross instrument movements and raises loss. These gaps are far smaller than in the open-loop experiment. That finding is important, but it is not evidence that feedback Nash is always benign: the recursive experiment also changes the private-sector timing from (1)–(2) to (19)–(20). A clean open-loop-versus-feedback comparison requires both concepts to be solved under the same rational-expectations closure.

Table 4: Recursive equilibrium and delegated adjustment coefficients

Regime	λ_M	λ_F	$\rho(A_{cl})$	demand loss	supply loss
Social regulator	—	—	0.9371	0.20278	0.70775
Nash, no inertia	0	0	0.9384	0.20565	0.70964
Feedback Nash	0.050	0.040	0.9383	0.20378	0.70816
Social institutional design	0.050	0.115	0.9382	0.20333	0.70802

6.7 The first-stage strategic-investment game

Figure 10 displays the institutional game. The monetary and fiscal best-response schedules cross at $(\lambda_M, \lambda_F) = (0.05, 0.04)$, exactly the primitive adjustment weights. The private stage-one choice therefore does not create an inertia arms race in this calibration. This is a substantive negative result: adjustment costs make current policy an investment in a strategic state, but that fact alone is not sufficient to generate privately excessive investment in commitment.

The social institutional choice is $(0.05, 0.115)$. Relative to feedback Nash, it leaves monetary delegation unchanged and almost triples delegated fiscal inertia. This reduces the large initial fiscal response to a demand shock from 0.155 to 0.106 and lowers loss in both reported shock experiments. The coordination failure in the first stage is therefore under-provision of fiscal commitment: the fiscal authority does not internalise how a smoother fiscal rule improves the monetary–fiscal policy mix. This result is conditional on the mandate weights, implementation cost, shock distribution, and recursive timing; the code reports the full objective surface so those assumptions can be varied rather than hidden.

7 Is the current approach sufficient?

The open-loop approach alone is not sufficient for the mechanism in the motivating draft. The combined project now supplies both the open-loop commitment benchmark and a recursive two-stage game. It demonstrates that mandate differences can create policy-mix externalities, keeps social primitives fixed, distinguishes steady state from shocks, and makes lagged policy a state. Three boundaries nevertheless remain.

First, a known deterministic shock path eliminates the arrival of new information. Second, open-loop paths prevent policies from responding to off-equilibrium states, whereas debt and lagged instruments are precisely the states through which strategic commitment operates. Third, open-loop leadership conflates leadership with full path commitment; feedback Stackelberg remains to be solved. Third, the implemented recursive block is semi-structural.

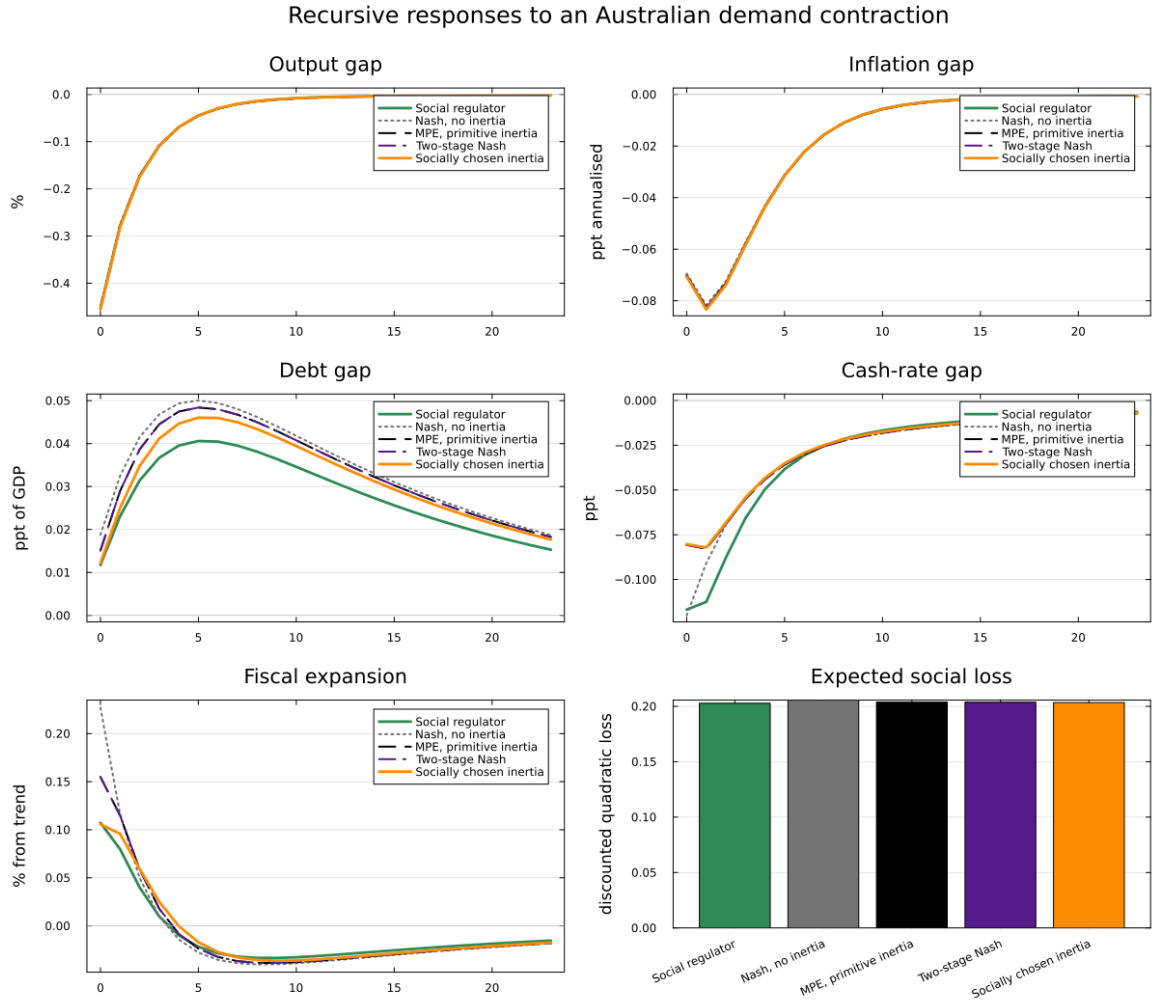


Figure 8: Recursive feedback responses to an Australian demand contraction. The two-stage Nash coincides with the primitive-inertia MPE in the baseline.

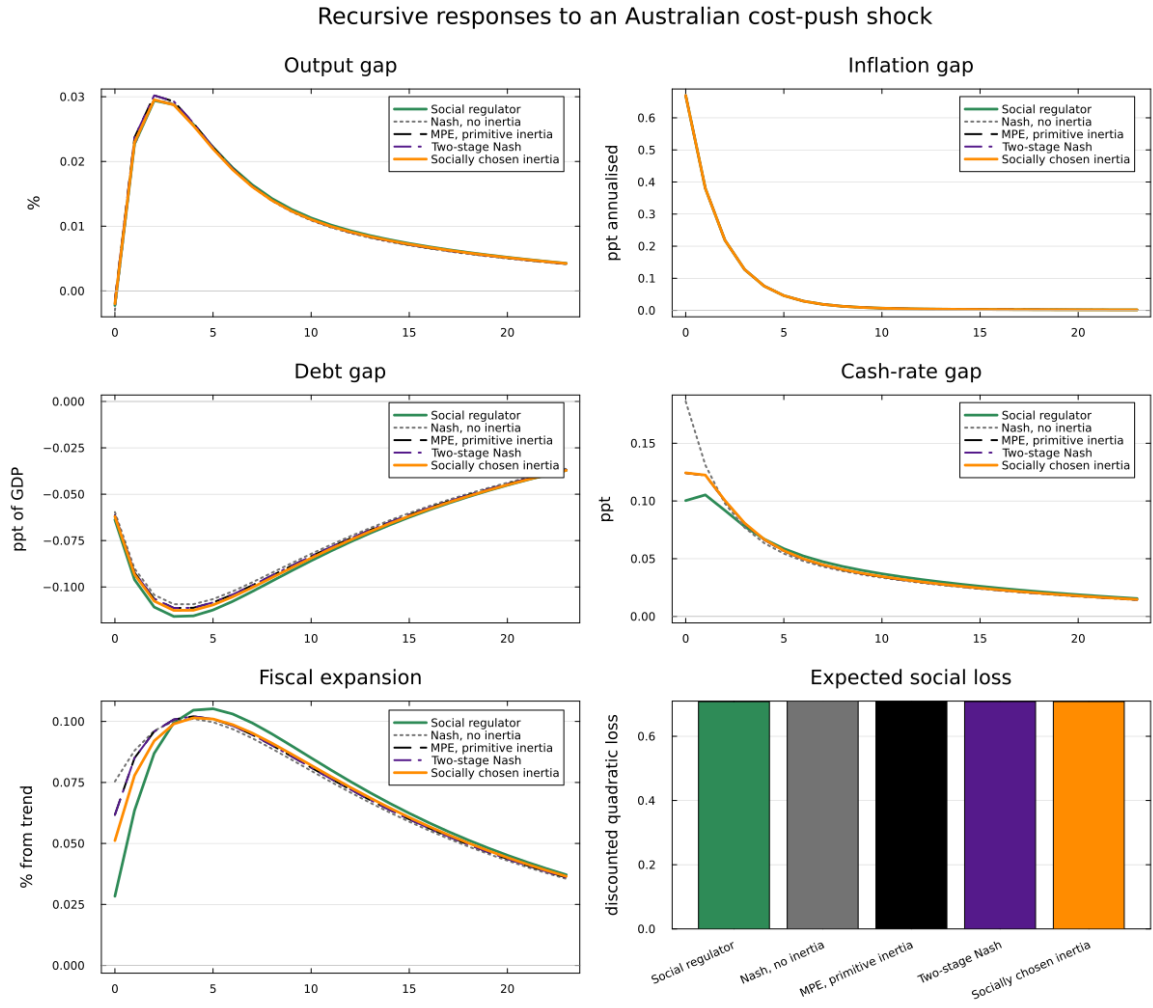


Figure 9: Recursive feedback responses to an Australian cost-push shock.

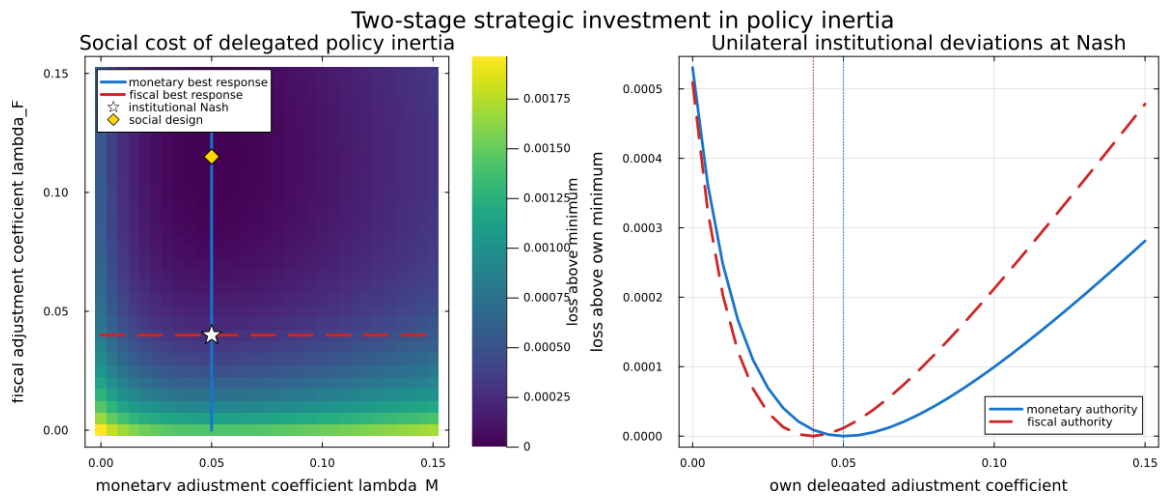


Figure 10: The two-stage game in delegated adjustment coefficients. Lines are the authorities' discrete best responses; the star is institutional Nash and the diamond is the social institutional design.

It makes timing and persistence transparent, but it does not yet solve private rational expectations jointly with the feedback rules. Nominal rigidity makes that remaining omission especially material after a cost-push shock, because expected future policy affects current inflation and output.

Table 5: Role of the alternative solution concepts

Concept	What it identifies	Limitation or role
Open-loop Nash	Policy-mix externality when complete paths are chosen after a known shock	Useful commitment benchmark; no reoptimisation after new states
Open-loop Stackelberg	Value of internalising the follower's complete-path reaction	Leadership and full-path commitment are bundled together
Feedback/MPE Nash	Discretionary, state-contingent interaction under recurring shocks	Appropriate main equilibrium for debt and lagged-policy states
Feedback Stackelberg	State-contingent strategic leadership	Required to test whether the two leadership regimes genuinely differ
Commitment meta-game	Ex ante choice of adjustment costs, delegation, or rules	Implemented here; a prisoner's dilemma is possible but does not arise at the baseline

8 What the recursive experiment establishes

The implemented experiment completes the first three necessary steps:

1. It solves the cooperative regulator and stabilising feedback Nash equilibrium, verifies convergence and closed-loop stability, and reports the full state-contingent rule coefficients.
2. It shows directly that removing delegated adjustment costs removes the own-lag terms from both policy rules. Policy history is therefore an active state rather than a verbal interpretation attached after the calculation.
3. It solves (17) for both authorities and for a social institutional designer. The difference between $(0.05, 0.04)$ and $(0.05, 0.115)$ is the measured first-stage coordination failure.

The next extension should solve the same feedback and open-loop concepts under one forward-looking rational-expectations block. That requires finding private expectation mappings jointly with the coupled policy Riccati equations. It should then add feedback Stackelberg equilibria, persistent estimated shock processes and covariance, and nominal government liabilities for a full fiscal dominance experiment. These additions would identify whether the small recursive welfare gaps are due to feedback discretion itself or to the current semi-structural timing.

The completed design already lets the steady-state and shock insights reinforce rather than contradict each other. The zero-gap deterministic steady state is a common reference point. Institutions matter ex ante because they determine feedback rules used away from that point. Demand and cost-push shocks activate different strategic margins, while the commitment stage determines whether the authorities choose institutions that mitigate those margins.

9 Conclusion

Different relative inflation and output weights are sufficient to generate a substantive monetary–fiscal game once shocks create a macroeconomic trade-off. They need not create steady-state conflict. In the Australian calibration, the no-shock steady state is common to all regimes, but shocks activate the strategic interaction. The open-loop benchmark displays a strong offsetting policy game near its reaction boundary. The recursive MPE is stable and history dependent, yet its welfare gaps are modest under the semi-structural Australian timing.

Most importantly, the two-stage game now distinguishes adjustment costs as true primitives from adjustment coefficients delegated to future policymakers. The decentralised institutional equilibrium retains the primitive coefficients; the social designer instead selects substantially greater fiscal inertia. The baseline concern is therefore not excessive strategic investment in commitment but too little coordinated fiscal commitment. This negative result disciplines the theory: strategic-state effects make a commitment game possible, but do not by themselves guarantee a prisoner’s dilemma. A common forward-looking private-sector closure and feedback Stackelberg solutions are the remaining steps needed to determine whether stronger strategic complementarity overturns that conclusion.

A Reproducibility

All calculations and figures are produced by the Julia project accompanying this paper. From the project directory, run

```
julia --project=. -e "using Pkg; Pkg.instantiate(; update_registry=false)"
julia --project=. run_model.jl
julia --project=. test/runtests.jl
julia --project=. build_tex.jl
```

The first command installs exact package versions from the manifest. The model runner writes calibration primitives, summary tables, the zero-shock audit, the shock-scaling experiment, recursive policy rules, the complete institutional objective grid, and all figures to `output/`. The final command invokes the Tectonic binary supplied by Julia’s artifact system and creates `PAPER.pdf` from this native L^AT_EX source.

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